QP Code:

**MAHATMA GANDHI UNIVERSITY, KOTTAYAM**
**MGU-UGP (HONOURS) REGULAR EXAMINATION OCTOBER 2025**
**THIRD SEMESTER**
**Discipline Specific Elective (DSE) - MG3DSEPHY204**
**INTRODUCTION TO MATERIALS SCIENCE**
(2024 ADMISSION ONWARDS)

**Duration:** 1.5 Hours **Maximum Marks:** 50

## PART A

**Short Answer Questions**
*Answer any 10 questions.*
*Each question carries 2 mark.*

1. Classify materials based on the types of bonding. [K]/[CO1]
2. Define ionic and covalent bonding with one example each. [U]/[CO1]
3. What is meant by secondary bonding? Give an example. [U]/[CO1]
4. Write down two differences between crystalline and non-crystalline solids. [K]/[CO1]
5. What is Bravais lattice and give any two examples. [K]/[CO1]
6. Define Unit cell. [U]/[CO1]
7. What is the physical significance of the Gibbs phase rule? [K]/[CO4]
8. Give one example each for unary and binary phase diagrams. [K]/[CO4]
9. Define elastic and plastic deformation. [U]/[CO6]
10. What is creep? Mention one factor affecting it. [K]/[CO6]
11. Explain the term ductility. [U]/[CO6]
12. What are point defects? Give two examples. [K]/[CO3]
13. What is toughness of a material? [U]/[CO6]
14. Define corrosion. Mention one way to prevent it. [K]/[CO6]

## PART B

**Short Essay Questions**
*Answer any 6 questions.*
*Each question carries 5 marks.*

15. Discuss the different types of atomic bonding in solids with suitable diagrams. [U]/[CO1]
16. Explain the concept of Miller indices and describe the procedure for finding the Miller indices of a plane. [A]/[CO1]
17. Draw and explain a typical binary phase diagram. Mention its important features. [U]/[CO4]
18. Explain phase transformations and write a note on the kinetics of transformation. [K]/[CO5]
19. Discuss elastic, inelastic and viscoelastic behaviour of materials with examples. [U]/[CO6]

20. What are different imperfections in crystals? Explain point and line defects with neat sketches. [U]/[CO3]
21. Write short notes on (a) Slip and Twinning, (b) Factors affecting fatigue in metals. [K]/[CO6]
22. Explain the mechanism of oxidation and corrosion in metals and describe one method to prevent it. [A]/[CO6]